Are Harris County Drivers the Worst in Texas?

An Editable Beamer Presentation

Fredy Guevara

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Senior project comparing selected Texas counties using crash rates, confidence intervals, and regression.

Selected counties: Harris, Travis, Bexar, Montgomery, Hays, Medina, Hidalgo, Hale, and Lipscomb.

Research Question

Main Question

Is Harris County the county with the worst drivers among the counties in this sample?

Working Idea

This project treats “worst drivers” as a question of:

- crash frequency relative to population
- fatal crash risk
- risky driving patterns such as speeding, DUI, and distraction

Why Total Crashes Are Not Enough

A large county will usually have more crashes simply because it has more people and more traffic.

That means total crashes alone can be misleading.

Better comparison measures

- crash rate per 100,000 people
- fatal crash rate per 100,000 people
- proportions involving speeding, DUI, and distracted driving

Counties in the Study

Table 1: Counties included in the analysis

County
Harris
Travis
Bexar
Montgomery
Hays
Medina
Hidalgo
Hale
Lipscomb

Creating New Variables

To compare counties fairly, several derived variables were created.

$$\text{Crash Rate} = \frac{\text{Total Crashes}}{\text{Population}} \times 100{,}000$$

$$\text{Fatal Crash Rate} = \frac{\text{Fatal Crashes}}{\text{Population}} \times 100{,}000$$

$$\text{Speed Percent} = \frac{\text{Speed Crashes}}{\text{Total Crashes}}$$

$$\text{DUI Percent} = \frac{\text{DUI Crashes}}{\text{Total Crashes}}$$

$$\text{Distracted Percent} = \frac{\text{Distracted Crashes}}{\text{Total Crashes}}$$

Severity Measure

To capture how dangerous crashes are, I also used:

$$\text{Severity Percent} = \frac{\text{Fatal Crashes} + \text{Serious Crashes}}{\text{Total Crashes}}$$

This measure helps distinguish between counties that have many crashes and counties that have more severe crashes.

Summary of Derived Variables

Table 2: Derived variables used in the analysis

County	Crash_Rate	Fatal_Rate	SpeedPercent	DUIPercent	DistractedPercent	SeverityPercent
Harris	2301.6684	10.9115	0.3240	0.0291	0.0855	0.2503
Bexar	2238.5467	9.4576	0.1549	0.0341	0.3660	0.0202
Hidalgo	1788.1609	6.1397	0.3359	0.0453	0.0947	0.0211
Montgomery	1592.3210	7.6058	0.3705	0.0393	0.1459	0.0291
Medina	1475.8063	19.4185	0.2344	0.0550	0.2285	0.0514
Hale	1437.0165	12.5778	0.2735	0.0547	0.2363	0.0416
Hays	1221.6796	8.8183	0.3115	0.0666	0.2526	0.0455
Travis	1137.0393	9.9577	0.1920	0.0745	0.3235	0.0410
Lipscomb	910.7468	36.4299	0.2000	0.0000	0.0800	0.0400

Ranking Counties by Crash Rate

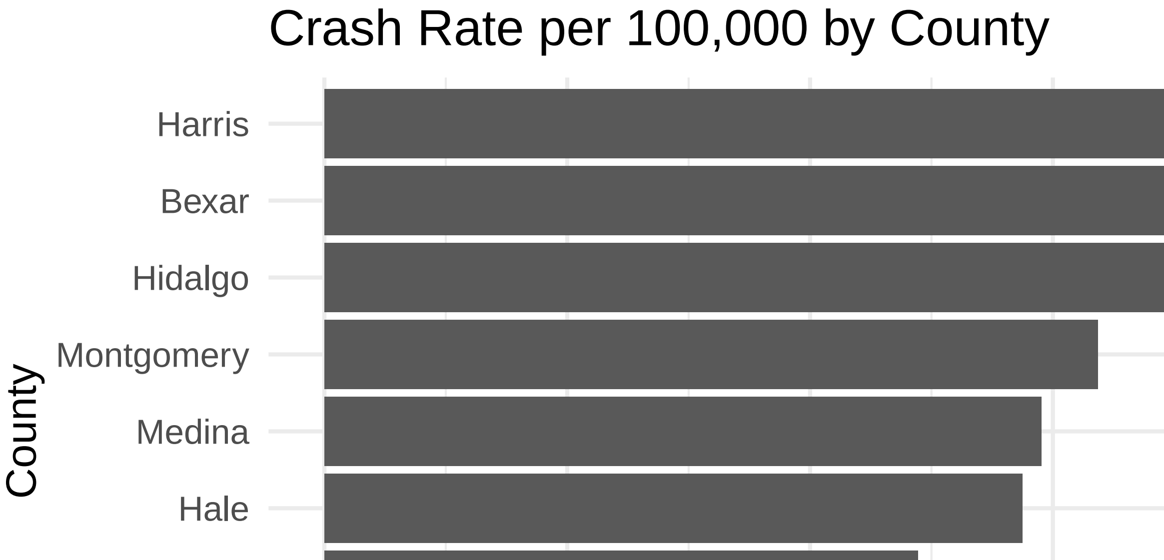
Table 3: Counties ranked by crash rate per 100,000

County	Crash_Rate	Total_Crashes	Population
Harris	2301.67	115173	5003892
Bexar	2238.55	48522	2167567
Hidalgo	1788.16	16601	928384
Montgomery	1592.32	12352	775723
Medina	1475.81	836	56647
Hale	1437.02	457	31802
Hays	1221.68	3602	294840
Travis	1137.04	15872	1395906
Lipscomb	910.75	25	2745

Brief interpretation

Ranking by crash rate is more meaningful than ranking by total crashes because it adjusts for county size.

Crash Rate by County



Ranking Counties by Fatal Crash Rate

Table 4: Counties ranked by fatal crash rate per 100,000

County	Fatal_Rate	Fatal_Crashes	Population
Lipscomb	36.43	1	2745
Medina	19.42	11	56647
Hale	12.58	4	31802
Harris	10.91	546	5003892
Travis	9.96	139	1395906
Bexar	9.46	205	2167567
Hays	8.82	26	294840
Montgomery	7.61	59	775723
Hidalgo	6.14	57	928384

Brief interpretation

Fatal crash rate focuses on severity rather than frequency, so it can highlight counties where crashes are less common but more dangerous.

Risk Factor Table

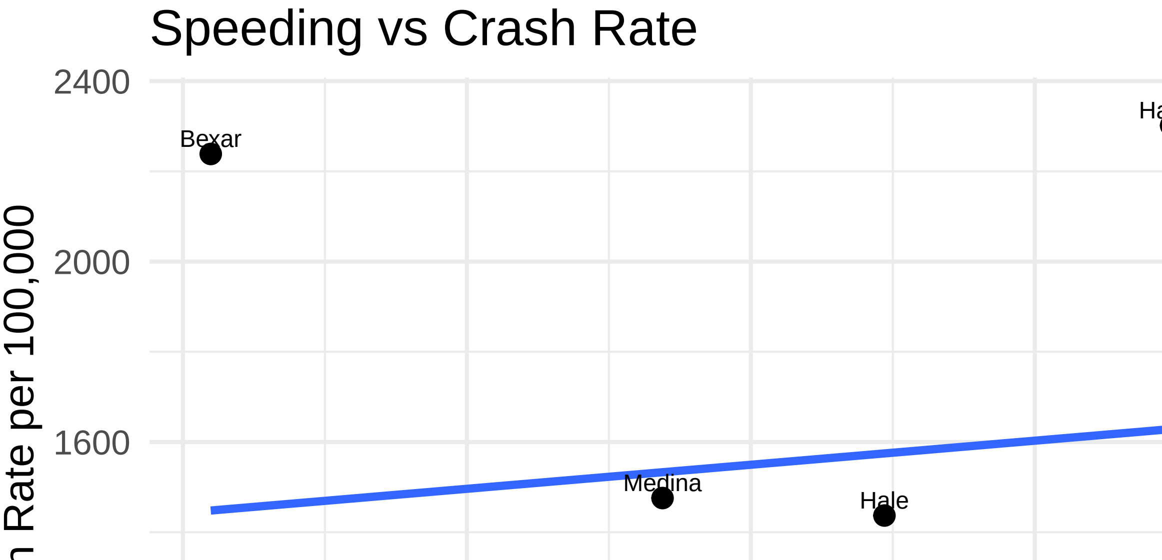
Table 5: Risk-factor proportions by county

County	SpeedPercent	DUIPercent	DistractedPercent	SeverityPercent
Montgomery	0.3705	0.0393	0.1459	0.0291
Hidalgo	0.3359	0.0453	0.0947	0.0211
Harris	0.3240	0.0291	0.0855	0.2503
Hays	0.3115	0.0666	0.2526	0.0455
Hale	0.2735	0.0547	0.2363	0.0416
Medina	0.2344	0.0550	0.2285	0.0514
Lipscomb	0.2000	0.0000	0.0800	0.0400
Travis	0.1920	0.0745	0.3235	0.0410
Bexar	0.1549	0.0341	0.3660	0.0202

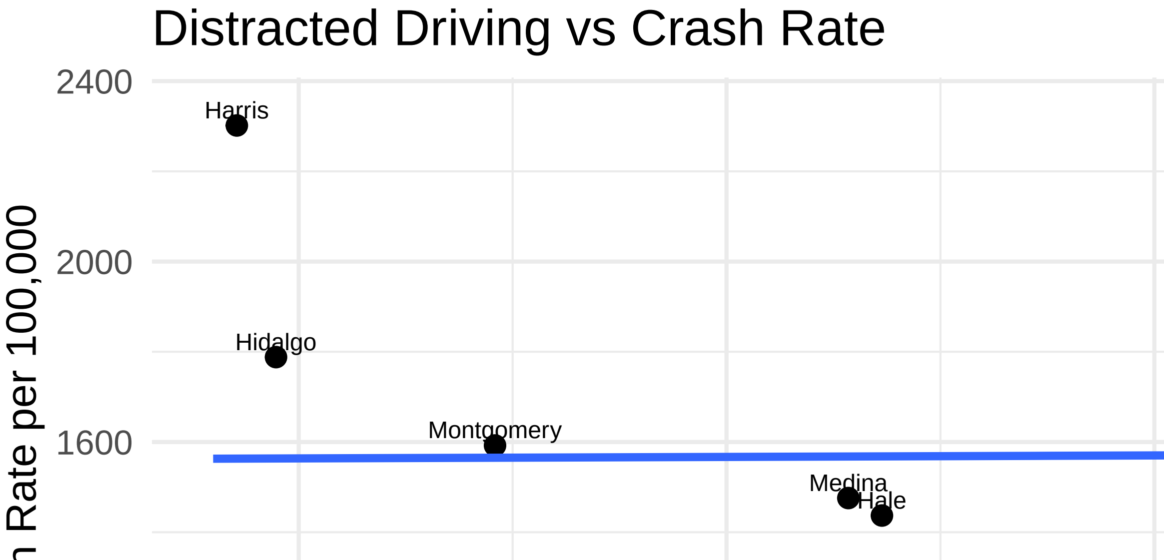
Brief interpretation

These percentages help explain why crash rates differ across counties. High values suggest that risky behavior may contribute to county-level differences.

Speeding vs Crash Rate



Distracted Driving vs Crash Rate



Confidence Interval Setup

To estimate the true county crash proportion, I used:

$$\hat{p} = \frac{\text{Total Crashes}}{\text{Population}}$$

$$\hat{p} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

where $z = 1.96$ for a 95% confidence level.

Confidence Intervals: Harris vs Montgomery

Table 6: 95% confidence intervals for crash rate per 100,000

County	Lower_CI	Upper_CI	Crash_Rate
Harris	2288.53	2314.81	2301.67
Montgomery	1564.46	1620.18	1592.32

Brief interpretation

If the two intervals overlap a lot, then the difference between Harris and Montgomery may not be statistically meaningful.

Regression Model

The regression model used was:

$$\text{Crash Rate} = \beta_0 + \beta_1(\text{SpeedPercent}) + \beta_2(\text{DUIPercent}) + \beta_3(\text{DistractedPercent}) + \varepsilon$$

This model tests whether crash rate is associated with speeding, DUI, and distracted-driving shares.

Regression Output

Table 7: Regression coefficients

term	estimate	std.error	statistic	p.value
(Intercept)	84.007	1372.552	0.0612	0.9536
SpeedPercent	5107.916	4239.003	1.2050	0.2821
DUIPercent	-16501.027	14148.702	-1.1663	0.2961
DistractedPercent	4237.233	3694.462	1.1469	0.3033

Model Fit

Table 8: Regression model fit

r.squared	adj.r.squared	sigma	statistic	p.value
0.25	-0.2	519.6527	0.5556	0.6667

Brief interpretation

The coefficients show the direction and size of each relationship. Smaller p-values suggest stronger evidence that a variable matters. The R^2 value shows how much of the variation in crash rates is explained by the model.

Main Takeaways

- Total crashes are not enough to identify the “worst drivers.”
- Crash rate and fatal crash rate provide fairer county comparisons.
- Risk-factor percentages help explain why counties differ.
- Harris County should be judged by rate-based measures, not just size.

Limitations

- The sample includes selected counties rather than all Texas counties.
- County-level analysis cannot identify individual driver behavior.
- Population-based crash rates are useful, but vehicle miles traveled would be an even stronger exposure measure.
- Small counties may produce unstable percentages because of low counts.

Conclusion

This project uses rates, confidence intervals, and regression to study whether Harris County stands out as having the worst drivers among the selected counties.

The final answer depends on whether Harris ranks highest after adjusting for population and whether its estimated crash rate is clearly different from comparable counties.